

Communication Protocol between KM5S Display and Controller

UART 9600 N 8 1

The display sends out the data to controller every 0.5 second. After receiving the right data, the controller sends back the running status data back to the display. If after 10 seconds no response from the controller, the display shows error 30.

Display->Controller Data Format

System Parameters Setting

0x3A+0x1A+0x53+0x07+0x02+0x40+0x01+0x0A+0x8F+0x32+0x7D+ 0xFF+0x01+0x0D+0x0A

Data definition:

Byte 1: 0x3A: Starting Byte, Fixed.

Byte 2: 0x1A: Controller Address, Fixed.

Byte 3: 0x53: 'S' means setting parameters, Fixed.

Byte 4: 0x07: Useful Data Length 7, Fixed.

Byte 5: 0x02: Pedal Sensor Parameters 1

bit7: Reserved

bit6: 0-Forward Pulse Duty Cycle<50%, 1-Forward Pulse Duty Cycle>50%, default value: 0

bit5~0: Output start after magnet numbers, default value: 2

Byte 6: 0x40: Pedal Sensor Parameters 2

PAS ratio: 0~255, default value: 6 magnets:128, 12 magnets:64.

Byte 7: 0x01: Speed sensor magnet numbers

bit7: Throttle limited by PAS levels, 0: No, 1:Yes

bit6: Throttle limited 6km/h, 0: No, 1:Yes

bit5~4: Start up Acceleration, 0:Strongest, 3:Slowest

bit3~0: Speed sensor magnets per circle

Byte 8: 0x0A: Cutoff voltage deviation

bit7: 0:Positive,1: Negative

bit6~0: Deviation, 1=0.1V

Byte 9: 0x8F: Cutoff voltage and Current Limit

bit7~6: Cutoff voltage base value 1: 21V 2: 31.5V 3: 42V

bit5~0: Current Limit x2

If the cutoff voltage is 30V, current Limit is 15A, this byte should be 0x9E, the previous byte should be 0x8F.

Byte 10: 0x32: Shake hands byte

Random data 0~63

Byte 11: 0x7D: Speed limit

bit7~3: Speed limit (km/h)-10

bit2~0: Wheel size ,0:16 inch, 1:18inch, 2:20inch, 3:22inch, 4:24inch, 5:26inch, 6:700c, 7:28inch

If the wheel size is 26 inch, speed limit is 25km/h, this byte should be 0x7D.

Byte 12: 0xFF: Checksum low byte

Accumulation (low byte) (not include start character 0x3A)

Byte 13: 0x01: Checksum high byte
Accumulation (high byte) (not include start character 0x3A)
Byte 14: 0x0D: Terminator 1, Fixed.
Byte 15: 0x0A: Terminator 2, Fixed.

Normal running

0x3A+0x1A+0x52+0x02+0xAB+0x80+0x99+0x01+0x0D+0x0A

Definition:

Byte 1: 0x3A: Starting Byte, Fixed.
Byte 2: 0x1A: Controller Address, Fixed.
Byte 3: 0x52: 'R' means running data, Fixed.
Byte 4: 0x02: Useful Data Length 2, Fixed.
Byte 5: 0xAB: Maximum PWM output 171 (Max.255)
Byte 6: 0x80: Status
bit7: Lights,0:Off, 1:On
bit6: Reserved
bit5: Low battery voltage, 0:No, 1:Yes
bit4: 6km/h,0: Off, 1: On
bit3: Reserved
bit2: Reserved
bit1: Reserved
bit0: Overspeed, 0:No, 1:Yes

Byte 7: 0x99: Checksum low byte
Accumulation (low byte) (not include start character 0x3A)
Byte 8: 0x01: Checksum high byte
Accumulation (high byte) (not include start character 0x3A)

Byte 9: 0x0D: Terminator 1, Fixed.
Byte 10: 0x0A: Terminator 2, Fixed.

Controller->Display Data Format

0x3A+0x1A +0x53+0x05+0x00+0x2D+0x0D+0xAC+0x00+0x58+0x01+0x0D+0x0A

Definition:

Byte 1: 0x3A: Starting Byte, Fixed.
Byte 2: 0x1A: Controller Address, Fixed.
Byte 3: 0x53: If the display sends 0x53, then reply 0x53, If the display sends 0x52, then reply 0x52.
Byte 4: 0x05: Useful Data Length 5, Fixed.
Byte 5: 0x00: Status:
bit7: Low battery voltage, 0:No, 1:Yes
bit6: Reserved
bit5~0: Reserved

Byte 6:0x2D: Current

Battery Output Current, $1=1/3A$, if the current is 15A, this byte should be 0x2D.

Byte 7:0x0D:

Byte 8:0xAC: Wheel running time per circle (Unit: ms), high byte first, low byte. Maximum time: 3500ms, i.e. 0x0DAC

Byte 9: Fault codes:

0x00: No error

0x21: current exception

0x22: turn the exception

0x23: Motor phase

0x24: motor abnormalities Hall

0x25: brake abnormal

Byte 10: 0x58:Checksum low byte

Accumulation (low byte) (not include start character 0x3A)

Byte 11: 0x01:

Checksum high byte

Accumulation (high byte) (not include start character 0x3A)

Byte 12: 0x0D: Terminator 1, Fixed.

Byte 13: 0x0A: Terminator 2, Fixed.

Notes: In KM5S protocol, the display does not send PAS level to controller, but send maximum PWM output. When in 6km/h walk assistance mode, this byte sends 51.